

Arshad Arjunan Nair

Atmospheric scientist

nairarshad@gmail.com | Website | Google Scholar | ORCID: 0000-0003-2530-7757

Research areas: Aerosol microphysics; physics-informed machine learning; aerosol–cloud interactions; atmospheric composition; air pollution exposure and health.

Previous academic appointments

Appointed Dec 2025	Former Ramanujan Fellow Space Physics Laboratory, Vikram Sarabhai Space Centre, ISRO
Appointed May 2024	Research Scientist Atmospheric Sciences Research Center, University at Albany, SUNY
Aug 2021–May 2024	Postdoctoral Associate Atmospheric Sciences Research Center, University at Albany, SUNY
2016–2021	Doctoral research, University at Albany, SUNY ASRC Graduate Fellow (appointed Aug 2016); Research Project Assistant (appointed Aug 2017)
Mar 2015–May 2016	Research Assistant Indian Institute of Tropical Meteorology, Pune, India

Education

2021	Ph.D., Atmospheric Sciences University at Albany, SUNY Dissertation: The role of ammonia in atmospheric new particle formation and implications for cloud condensation nuclei. Advisor: Fangqun Yu.
2015	B.S.–M.S., Physics Indian Institute of Science Education and Research (IISER) Pune Thesis: Lightning distribution during active and break monsoon periods over South Asia. Advisor: Adarsh Kumar Kamra.

Selected honors and fellowships

- **Ramanujan Fellowship**, Anusandhan National Research Foundation, Government of India (2025).
- **GeoCAFE Scholars Program**, Boston University and Harvard T.H. Chan schools of public health (2025).
- **Narayan R. Gokhale Distinguished Research Scholarship Award**, University at Albany (2023).
- **First Place, Chancellor Distinguished Ph.D. Graduate Dissertation Awards**, SUNY (2022 award cycle; announced January 2023).
- **Distinguished Doctoral Dissertation Award**, University at Albany (2021–2022).

Peer-reviewed publications

Author name in bold; † denotes equal contribution. Author lists are abbreviated for selected multi-author papers.

1. Yu, F., Luo, G., & **Nair, A. A.** (2026) Higher efficacy of SO₂ and accumulation mode-H₂SO₄ Stratospheric Aerosol Injection: Insights from CESM2 and GEOS-Chem with Advanced Particle Microphysics (APM), *Environmental Research Letters*, 21(4), 044006. 10.1088/1748-9326/ae3c38
2. Qi, Q., Xue, Y., Madani, N. A., Tangang, R. T., Yu, F., **Nair, A. A.**, Xiaobo Romeiko Xue, Luo, G., Brackett, I., Thorncroft, C. & Lin, S. (2025) Individual effects and interactions between ultrafine particles and extreme temperatures on hospital admissions of high burden diseases, *Environment International*, 197, 109348. 10.1016/j.envint.2025.109348
3. Qi, Q., Yu, F., **Nair, A. A.**, Lau, S.S., Luo, G., Mithu, I., Zhang, W., Li, S. & Lin, S. (2024). Hidden danger: The long-term effect of ultrafine particles on mortality and its sociodemographic disparities in New York State, *Journal of Hazardous Materials*, 471, 134317. 10.1016/j.jhazmat.2024.134317
4. Yu, F., Anderson, B.E., Pierce, J.R., Wong, A., **Nair, A. A.**, Luo, G. & Herb, J. (2024). On Nucleation Pathways and Particle Size Distribution Evolutions in Stratospheric Aircraft Exhaust Plumes with H₂SO₄ Enhancement, *Environmental Science & Technology*, 58(16), 6934–6944. 10.1021/acs.est.3c08408
5. Yu, F.[†], **Nair, A. A.**[†], Lauper, U., Luo, G., Herb, J., Morse, M. J., Savage, B., Zartarian, M., Wang, M., & Lin, S. (2024). Mysteriously rapid rise in Legionnaires' disease incidence correlates with declining atmospheric sulfur dioxide, *PNAS Nexus*, 3(3), pgae085. 10.1093/pnasnexus/pgae085. [†]Equal contribution.
6. **Nair, A. A.**, Lin, S., Luo, G., Ryan, I., Qi, Q., Deng, X., & Yu, F. (2023) Environmental exposure disparities in ultrafine particles and PM_{2.5} by urbanicity and socio-demographics in New York State, 2013–2020. *Environmental Research*, 239(2), 117246. 10.1016/j.envres.2023.117246
7. **Nair, A. A.**, Yu, F., Luo, G. (2023). The importance of ammonia for springtime atmospheric new particle formation and aerosol number abundance over the United States. *Science of the Total Environment*, 863, 160756. 10.1016/j.scitotenv.2022.160756
8. Yu, F., Luo, G., **Nair, A. A.**, Eastham, S., Williamson, C. J., Kupc, A., and Brock, C. A. (2023). Particle number concentrations and size distributions in the stratosphere: Implications of nucleation mechanisms and particle microphysics. *Atmospheric Chemistry and Physics*, 23(3), 1863–1877. 10.5194/acp-23-1863-2023
9. Yu, F., Luo, G., **Nair, A. A.**, Tsigaridis, K., Bauer, S. (2022). Use of machine learning to reduce uncertainties in particle number concentration and aerosol indirect radiative forcing predicted by climate models. *Geophysical Research Letters*, 49(16), e2022GL098551. 10.1029/2022GL098551
10. Lin, S., Ryan, I., Paul, S., Deng, X., Zhang, W., Luo, G., Dong, G., **Nair, A. A.**, & Yu, F. (2022). Particle surface area, ultrafine particle number concentration, and cardiovascular hospitalizations. *Environmental Pollution*, 310, 119795. 10.1016/j.envpol.2022.119795
11. Deng, X., Friedman, S., Ryan, I., ... , **Nair, A. A.**, ... & Lin, S. (2022). The independent and synergistic impacts of power outages and floods on hospital admissions for multiple diseases. *Science of the Total Environment*, 828, 154305. 10.1016/j.scitotenv.2022.154305
12. **Nair, A. A.**, Yu, F., Campuzano-Jost, P., DeMott, P. J., Levin, E. J. T., ... & Peng, Q. (2021). Machine learning uncovers aerosol size information from chemistry and meteorology to quantify potential cloud-forming particles. *Geophysical Research Letters*, 48, e2021GL094133. 10.1029/2021GL094133
13. Mao, J., Zhang, Y., Yu F., **Nair, A. A.**, Yu, Q., ... & Chen, L. (2021) On the ship particle number emission index: Size-resolved microphysics and key controlling parameters. *Journal of Geophysical Research: Atmospheres*, 126, e2020JD034427. 10.1029/2020JD034427

14. **Nair, A. A.** & Yu, F. (2020). Using machine learning to derive cloud condensation nuclei number concentrations from commonly available measurements. *Atmospheric Chemistry and Physics*, 20(21), 12853–12869. [10.5194/acp-20-12853-2020](https://doi.org/10.5194/acp-20-12853-2020)
15. **Nair, A. A.** & Yu, F. (2020). Quantification of atmospheric ammonia concentrations: A review of its measurement and modeling. *Atmosphere*, 11(10), 1092. [10.3390/atmos11101092](https://doi.org/10.3390/atmos11101092)
16. Yu, F., Luo, G., **Nair, A. A.**, Schwab, J. J., Sherman, J. P. & Zhang, Y. (2020). Wintertime new particle formation and its contribution to cloud condensation nuclei in the Northeastern United States. *Atmospheric Chemistry and Physics*, 20(4), 2591–2601. [10.5194/acp-20-2591-2020](https://doi.org/10.5194/acp-20-2591-2020)
17. **Nair, A. A.**, Yu, F. & Luo, G. (2019). Spatiotemporal variations of atmospheric ammonia concentrations over the United States: Comprehensive model-observation comparison. *Journal of Geophysical Research: Atmospheres*, 124(12), 6571–6582. [10.1029/2018JD030057](https://doi.org/10.1029/2018JD030057)
18. Yu, F., **Nair, A. A.** & Luo, G. (2018). Long-term trend of gaseous ammonia over the United States: Modeling and comparison with observations. *Journal of Geophysical Research: Atmospheres*, 123(15), 8315–8325. [10.1029/2018JD028412](https://doi.org/10.1029/2018JD028412)
19. Kamra, A. K. & **Nair, A. A.** (2015). The impact of the Western Ghats on lightning activity on the western coast of India. *Atmospheric Research*, 160, 82–90. [10.1016/j.atmosres.2015.03.006](https://doi.org/10.1016/j.atmosres.2015.03.006)

Research funding and project experience

Past fellowship award

Physics-informed AI for aerosol microphysics and climate–health linkages: machine learning to enhance Earth system modeling and analysis for climate resiliency.

Ramanujan Fellowship, Government of India, awarded 2025. **Role during fellowship: PI.** Fellowship appointment concluded.

Past roles on collaborative awards

Assessing climate and air-quality exposure and financial impacts at the census tract level. New York State Department of Financial Services. **Previous role: Investigator.** Project grant period: August 2025–July 2028. Total project award: \$1,611,585.

Evaluating short-/long-term impacts and exposure sources of ultrafine particles on multiple health outcomes in New York State by using high resolution pollutant simulations. NYSEDA. **Previous role: Named investigator.** Project grant period: October 2019–October 2024. Total project award: \$609,851.

Past supported project participation

Toward improved understanding of fundamental processes controlling the size-resolved properties of particles in the stratosphere. National Science Foundation. **Previous role: Postdoctoral researcher / other personnel.** Project grant period: December 2023–November 2026. Total project award: \$599,298.

Teaching and mentoring

Graduate course modules, University at Albany

Spring 2023	Aerosol exposure and COVID-19 Module in EHS 545: Global Climate Change, Extreme Weather & Public Health, Department of Environmental Health Sciences
Fall 2022	Machine learning and aerosol microphysics Module in ATM 515: Aerosol Physics, Department of Atmospheric and Environmental Sciences

Mentoring

- Co-guided a master's student's summer research on Bay of Bengal lightning climatology using 18 years of satellite observations, Indian Institute of Tropical Meteorology (2015).
- Mentor, Science and Technology Entry Program / College Overview and Research Experience (STEP/CORE), New York State Education Department.
- Mentor, Project SHORT; mentor, Wonderland Literacy Center, India.

Selected professional service

University at Albany

- Senate Council on Research, 2024–2026; research centers committee; Faculty Research Award Program Category A committee.
- University Research Data and Security Compliance Council, 2025–2026.
- Research Faculty Search Committee, Atmospheric Sciences Research Center, 2024.

Research community

- Proposal reviewer: National Science Foundation, NASA, and Belgian Science Policy Office.
- Journal reviewer (selected): *Science*; *Atmospheric Chemistry and Physics*; *Environmental Science & Technology*; *Environmental Science & Technology Letters*; *Environmental Research Letters*; *Environment International*; *Atmospheric Research*.
- Guest editor: *Atmosphere* and *International Journal of Environmental Research and Public Health*.
- Student presentation competition judge, American Meteorological Society Atmospheric Chemistry Symposium, 2023.

Selected presentations

Jan 2026 Physics-informed machine learning for aerosol microphysics: improving confidence in how atmospheric particles shape climate and health.

Invited colloquium, Department of Physics and Astronomy, Appalachian State University.

Jun 2025 Physics-informed machine learning parameterizations of aerosol microphysics for chemistry and climate models.

Gordon Research Conference on Machine Learning for Actionable Climate Science, Smithfield, RI.

Feb 2023 The role of ammonia in atmospheric new particle formation and implications for cloud condensation nuclei.

Invited presentation, Academic Affairs Committee, SUNY Board of Trustees.